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Authorship Attribution in the Era of LLMs: Problems, Methodologies, and Challenges

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ABSTRACT

Accurate attribution of authorship is crucial for maintaining the integrity of digital content, improving forensic investigations, and mitigating the risks of misinformation and plagiarism. Addressing the imperative need for proper authorship attribution is essential to uphold the credibility and accountability of authentic authorship. The rapid advancements of Large Language Models (LLMs) have blurred the lines between human and machine authorship, posing significant challenges for traditional methods. We present a comprehensive literature review that examines the latest research on authorship attribution in the era of LLMs. This survey systematically explores the landscape of this field by categorizing four representative problems: (1) Human-written Text Attribution; (2) LLM-generated Text Detection; (3) LLM-generated Text Attribution; and (4) Human-LLM Co-authored Text Attribution. We also discuss the challenges related to ensuring the generalization and explainability of authorship attribution methods. Generalization requires the ability to generalize across various domains, while explainability emphasizes providing transparent and understandable insights into the decisions made by these models. By evaluating the strengths and limitations of existing methods and benchmarks, we identify key open problems and future research directions in this field. This literature review serves a roadmap for researchers and practitioners interested in understanding the state of the art in this rapidly evolving field. Additional resources and a curated list of papers are available and regularly updated at https://llm-authorship.github.io/.

1 INTRODUCTION

Authorship Attribution (AA) is the process of determining the author of a particular piece of writing, which has significant real-world applications across various domains. In forensic investigations, authorship attribution plays a crucial role in solving murder cases disguised as suicides (Chaski, 2005, Grant, 2020), tracking terrorist threats (Cafiero and Camps, 2023, Winter, 2019), and aiding general criminal investigations (Argamon, 2018, Belvisi et al., 2020, Koppel et al., 2008). In the digital realm, authorship attribution helps safeguard the integrity of content by preventing deceptive social media activities (Hazell, 2023), detecting account compromises (Barbon et al., 2017), and linking user profiles across various social networks (Shu et al., 2017, Sinnott and Wang, 2021). Additionally, authorship attribution techniques are instrumental in combating misinformation (Chen and Shu, 2024a,b, Hanley and Durumeric, 2024, Shu et al., 2020, Stiff and Johansson, 2022), protecting intellectual property rights (Meyer zu Eissen et al., 2007, Stamatatos and Koppel, 2011), and identifying fraudulent practices such as fake reviews (Afroz et al., 2012, Ott et al., 2011).

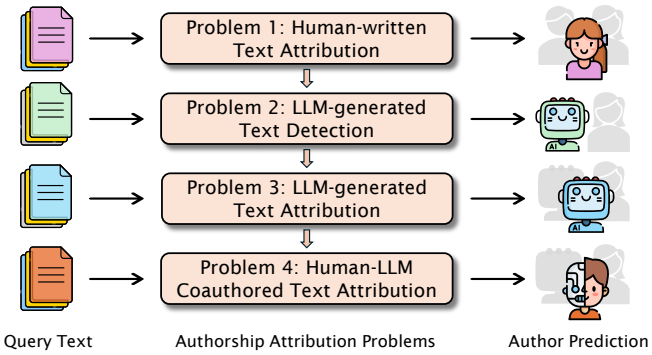

Figure 1: Four representative Problems in Authorship Attribution: (1) Human-written Text Attribution, which involves attributing an unknown text to its human authors; (2) LLM-generated Text Detection, which focuses on detecting whether a text has been generated by LLMs; (3) LLM-generated Text Attribution, aimed at identifying the specific LLM or human responsible for a given text; (4) Human-LLM Co-authored Text Attribution, which classifies a text as human-written, LLM-generated, or a combination of both. *These problems become progressively more complex, as indicated by the arrows.*

The development of large language models (LLMs) has revolutionized text generation, offering numerous benefits but also raising significant concerns about text authenticity and originality (Goldstein et al., 2023). The advent of LLMs has complicated authorship attribution, making it increasingly difficult to distinguish between LLM-generated texts and humans-written texts (Clark et al., 2021, Sadasivan et al., 2023). Identifying LLM-generated texts is challenging even for human experts, let alone traditional authorship attribution methods (Gao et al., 2022, Liu et al., 2023a). This inability to distinguish between human and machine-generated content undermines the integrity of authorship, complicates legal and ethical responsibilities, and threatens the credibility of digital content and the safety of online space (Solaiman et al., 2023, Vidgen et al., 2024).

Over the past few decades, authorship attribution has experienced significant advancements due to the development of natural language analysis tools and innovative methods for text representation learning. Traditionally, authorship attribution relied on stylometry, which analyzes an individual's unique writing style through feature engineering to capture linguistic characteristics (Lagutina et al., 2019). The emergence of machine learning algorithms capable of handling high-dimensional data has enabled the

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creation of more expressive representations. In recent years, there has been a shift towards extracting text embeddings using pre-trained language models (Fabien et al., 2020). These approaches, while offering higher performance, often sacrifice explainability for accuracy (Rivera-Soto et al., 2021). More recently, researchers have begun utilizing LLMs to extract features in conjunction with machine learning classifiers or to conduct end-to-end reasoning for authorship attribution (Huang et al., 2024, Patel et al., 2023).

The rapid advancements in LLMs have significantly improved text generation, producing outputs that rival human writing in fluency and coherence. This progress underscores the imperative need to distinguish between human-written text, LLM-generated text, or a combination of both. As illustrated in Figure 1, authorship attribution can be systematically categorized into four representative problems: attributing unknown texts to human authors, detecting LLM-generated texts, identifying the specific LLM or human responsible for a text, and classifying texts as human, machine, or human-LLM co-authored. Each task presents unique challenges that necessitate corresponding solutions. Researchers continually adapt and refine attribution methods, transitioning from human-authored texts to LLM-generated content, and navigating the complex interweaving in human-LLM co-authored works. As detection methods advance, adversarial attacks also evolve to bypass these measures, creating a continuous cycle of challenge and response in the quest to distinguish and disguise authorship (Dugan et al., 2024). Addressing these challenges will pave the way for more robust and reliable authorship attribution techniques.

Authorship attribution for both human and LLM-generated texts can be framed as either binary or multi-class classification. The LLM-generated text detection task simplifies the attribution problem by classifying each text as either originating from humans or LLMs (Jawahar et al., 2020, Mitchell et al., 2023, Pu et al., 2023a, Sadasivan et al., 2023). The majority of previous research on the automatic detection of machine-generated text has focused on binary classification (Jawahar et al., 2020, Mitchell et al., 2023). In the more challenging multi-class task, the goal is not only to differentiate between human and LLM-generated text but also to classify the text according to its specific source of generative models (Li et al., 2023b, Uchendu et al., 2021). Differences in LLM architectures, training methods, and generation techniques can influence the style of the generated texts (Munir et al., 2021). In the more complex human-LLM co-authoring problem, the goal is to distinguish texts authored by humans, LLMs, or combinations of both. Such nuanced detection provides deeper insights into the provenance of the text and is crucial for applications requiring detailed source attribution. Neural network-based detectors, generally outperform metric-based methods in both human authorship attribution and LLM-generated text detection problems (He et al., 2023, Zhang et al., 2024). However, these neural network approaches often offer less explainability compared to their metric-based counterparts.

This review serves as a valuable resource, comprehensively summarizing existing literature and highlighting the challenges and opportunities introduced by LLMs. We provide in-depth analysis of methodologies and datasets in this evolving field. The main contributions of this paper are as follows:

- We provide a timely overview to discuss the challenges and opportunities presented by LLMs in the field of authorship attribution. By systematically categorizing authorship attribution into four representative problems and balancing problem complexity with practicality, we reveal insights into the evolving filed of authorship attribution in the era of LLMs.
- We offer a comprehensive comparison of state-of-the-art methodologies, datasets, benchmarks, and commercial tools used in authorship attribution. This analysis not only improves the understanding of authorship attribution but also provides a valuable resource for researchers and practitioners to use as guidelines for approaching this direction.
- We discuss open issues and provide future directions by considering crucial aspects such as generalization, explainability, and interdisciplinary perspectives. We also discuss the broader implications of authorship attribution in real-world applications. This holistic approach ensures that authorship attribution not only yields accurate results but also provides insights that are explainable and socially relevant.

The remainder of this survey is organized as follows. In Section 2, we examine human authorship attribution, starting with a problem definition and moving through various methodologies, followed by a discussion on challenges. In Section 3, we discuss LLM-generated text detection. In Section 4, we explore LLM-generated text attribution. Section 5 covers Human-LLM co-authored text attribution. Section 6 discusses resources and evaluation metrics, comparing benchmarks and datasets. Section 7 highlights opportunities and future directions. In Section 8, we discuss ethical and privacy concerns. Finally, we conclude this survey in Section 9.

2 HUMAN AUTHORSHIP ATTRIBUTION

This section discusses authorship attribution of human-written texts. It covers various methodologies, including stylometry, machine learning, pre-trained language models, and LLM-based methods. Challenges such as limited data, evolving writing styles, and interpretability are also addressed.

2.1 Problem Definition

Authorship attribution aims to identify the author of an unknown text from a set of known authors. This can be an open-class problem, where the true author might not be among the known authors, or a closed-class problem, where the true author is included in a finite set of authors (Andrews and Bishop, 2019, Stolerman et al., 2014). Authorship attribution methods are typically divided into classification-based methods for a small set of candidate authors and similarity-based ranking methods for larger numbers of authors (Huertas-Tato et al., 2022, Rivera-Soto et al., 2021). These techniques can also be adapted to related problems such as authorship verification and profiling. Authorship verification determines whether a piece of writing was written by a specific individual (Stamatatos, 2016), while profiling infers characteristics such as age or gender from the author's writing style (Argamon et al., 2009).

2.2 Methodologies

We explore the evolution of methods used to analyze human-written text, starting with stylometry. Over time, the focus has shifted to using text embeddings. Recently, the integration of LLMs marks further advancements in attributing authorship.

Stylometry Methods. Stylometry, the quantitative analysis of writing style, has evolved from its initial reliance on human expertise (Mosteller and Wallace, 1963) to computational methods (Lagutina et al., 2019, Neal et al., 2017). This discipline utilizes a variety of linguistic features to determine authorship (Holmes, 1994, Lagutina et al., 2019), positing that each author's unique style can be captured through quantifiable characteristics (Argamon et al., 2009). Key stylometric features include character and word frequencies (Sharma et al., 2018), parts of speech (Sundararajan and Woodard, 2018), punctuation, topics (Halvani and Graner, 2021, Potha and Stamatatos, 2019, Seroussi et al., 2014), and vocabulary richness. Important features can be categorized into the following types: lexical, syntactic, semantic, structural, and content-specific (Rudman, 1997). Lexical features involve word choice and frequency; syntactic features pertain to sentence structure and grammar; semantic features explore the meaning and context of words; structural features relate to text organization; and content-specific features emphasize domain-specific terms (Bozkurt et al., 2007, Seroussi et al., 2014).

Machine Learning Methods. Machine learning approaches integrate stylometric features with classifiers such as logistic regression (Aborisade and Anwar, 2018, Madigan et al., 2005a), Bayesian multinomial regression (Argamon et al., 2009), and support vector machines (SVM) (Bacciu et al., 2019). Before the widespread adoption of transformer-based models, multi-headed Recurrent Neural Networks (RNNs) (Bagnall, 2015), and Long Short-Term Memory (LSTMs) were utilized at both sentence and article levels (Qian et al., 2017). Convolutional Neural Networks (CNNs) were applied at various levels, including characters, words, and N-grams (Ruder et al., 2016, Shrestha et al., 2017). Moreover, syntax-augmented CNN models (Zhang et al., 2018), Convolutional Siamese Networks (Saedi and Dras, 2021), and attention-based Siamese Networks (Boenninghoff et al., 2019a) were explored. Additionally, combinations of convolutions and transformers have been employed to learn embeddings for comparison tasks (Andrews and Bishop, 2019).

Pre-trained Language Models. Pre-trained language models, especially BERT-based architectures (Devlin et al., 2019) such as BERT (Fabien et al., 2020, Ippolito et al., 2020, Manolache et al., 2021), Sentence-BERT (Rivera-Soto et al., 2021, Schlicht and de Paula, 2021), and RoBERTa (Huertas-Tato et al., 2022), have proven effective for learning authorship representation. These methods do not require hand-crafted features but require substantial training time and domain-specific labeled data, struggling with cross-domain generalization and explainability. Contrastive learning (Khosla et al., 2020) is often used with pre-trained language models to enhance stylistic representation by maximizing similarity between texts written by the same authors and minimizing it with texts from different authors (Huertas-Tato et al., 2022).

Barlas and Stamatatos (2020) found that BERT performed well with large vocabularies, outperforming multi-headed RNNs. Fabien et al. (2020) fine-tuned a BERT model for authorship attribution, showing that including additional stylometric and hybrid features in an ensemble model can improve performance. Rivera-Soto et al. (2021) concluded that topic diversity and dataset size are crucial for effective cross-domain transfer. Adaptation through style transfer has not resolved cross-domain issues (Boenninghoff et al., 2019b, Wegmann et al., 2022). Techniques like slanted triangular learning rates and gradual unfreezing can be used to avoid catastrophic forgetting during fine-tuning (Howard and Ruder, 2018).

LLM-based Methods. Despite advances in LLMs, their potential for authorship attribution remains underexplored. The natural language understanding capability allows LLMs to recognize nuances, styles, and patterns in language, which are crucial for distinguishing between authors. Authorship attribution is a complex reasoning task, and LLMs possess significant capabilities in reasoning and problem-solving, particularly in zero-shot learning within resource-limited domains (Kojima et al., 2022). They assist in feature extraction by identifying syntactic patterns, lexical choices, and grammatical structures essential for authorship attribution. Traditionally, LLMs have been employed mainly for auxiliary tasks such as feature extraction and data annotation (Patel et al., 2023). Notable examples include the use of GPT-3 for data annotation (Brown et al., 2020) and a T5 Encoder for learning authorship signatures. Beyond feature extraction, LLMs can also be directly utilized for end-to-end authorship attribution (Huang et al., 2024).

The incorporation of LLMs in authorship attribution addresses several limitations of traditional methods. Unlike BERT-based models, which require computationally expensive fine-tuning and large amounts of domain-specific data for optimal performance, LLMs can generalize across various domains without fine-tuning, thereby mitigating issues related to domain specificity (Barlas and Stamatatos, 2020). LLMs are also effective with shorter texts, reducing the necessity for long inputs to derive meaningful representations (Eder, 2015). Another key advantage of LLM-based approaches is their ability to provide natural language explanations for their predictions, enhancing transparency compared to hidden text embeddings (Huang et al., 2024). This versatility marks a step forward in overcoming challenges related to data, domain specificity, text length requirements, and explainability faced by earlier methods.

2.3 Open Challenges

Human authors exhibit a diverse range of writing styles influenced by genre, topic, context, and temporal changes. This variability complicates authorship attribution, necessitating the identification of consistent and unique stylistic markers. Further complexity arises from the presence of noise due to varying document sizes and languages, requiring algorithms to manage a broad spectrum of linguistic nuances. Short or low-quality texts, such as social media posts, often provide unreliable data, complicating accurate attribution (Eder, 2015, Theophilo et al., 2021). Collaborative texts, which blend multiple writing styles, further mask individual contributions and obscure distinct authorial signals (Dauber et al., 2017).

Traditional stylometric methods rely on human expertise and manually crafted features, whereas deep learning methods demand significant computational resources and extensive labeled data, with the risk of catastrophic forgetting (Ramasesh et al., 2022). Authorship attribution using LLMs also faces several challenges: their

effectiveness decreases with an increasing number of candidate authors due to context length constraints (Huang et al., 2024), and they can perpetuate biases from training data, resulting in inaccurate attributions for texts from marginalized groups and languages (Liang et al., 2023). Additionally, LLMs can be misused to generate content that conceals true authorship by mimicking others or using LLMs to alter their work.

3 LLM-GENERATED TEXT DETECTION

LLMs excel at generating fluent and coherent text, raising concerns about the authenticity and originality of authorship. Detecting LLM-generated text is crucial for several applications, including combating misinformation and disinformation on social media (Chen and Shu, 2024b, Gambini et al., 2022, Stiff and Johansson, 2022), identifying spam (Jindal and Liu, 2008), preventing phishing attacks (Hazell, 2023), identifying fake reviews (Salminen et al., 2022), and detecting machine-generated scientific papers (Liu et al., 2023a, Rodriguez et al., 2022). As a result, the detection of LLM-generated text has garnered significant attention (Kumarage and Liu, 2023, Tang et al., 2023, Wu et al., 2023b, Yang et al., 2023b).

3.1 Problem Definition

LLM-generated texts are included within the scope of machine-generated texts[1]. Machine-generated texts encompass any text produced by automated systems, including simpler language models or rule-based systems (Uchendu et al., 2021). This paper focuses specifically on LLM-generated texts. The task of detecting LLM-generated text involves identifying and distinguishing text created by LLMs from that written by humans. Typically, this task is approached as a binary classification problem (Fagni et al., 2021, Jawahar et al., 2020, Mitchell et al., 2023, Solaiman et al., 2019, Zellers et al., 2019).

3.2 Methodologies

Evaluating the quality of machine-generated excerpts has traditionally relied on human judgment, which is considered the gold standard for open-domain generation systems (Gehrmann et al., 2019, van der Lee et al., 2019). However, distinguishing between LLM-generated and human-written texts poses significant challenges for humans (Dugan et al., 2023). For example, untrained human reviewers often fail to distinguish GPT-3-generated text from human-written text, identifying it correctly only at a rate consistent with random chance (Clark et al., 2021). Liu et al. (2023a) found that even experienced faculty members and researchers could only achieve about a 50% success rate in identifying GPT-generated academic writings. In contrast, detection algorithms frequently outperform human reviewers in this task (Ippolito et al., 2020).

Chakraborty et al. (2023a) employed theoretical analysis to argue that detecting LLM-generated text is nearly always feasible with the collection of multiple samples, and they established precise sample complexity bounds for this detection. However, existing detectors and models for LLM-generated text are not yet fully reliable (Dugan et al., 2024, Sadasivan et al., 2023, Wang et al., 2023). Sadasivan et al. (2023) provided theoretical insights indicating that the detection problem is becoming increasingly difficult.

[1]Machine-generated texts is also referred to as machine-authored, AI-generated, neural-generated, deepfake text, neural text, or synthetic text.

LLM-generated text detectors can be categorized into metric-based and model-based methods (He et al., 2023), which are further divided into feature-based, neural network-based, zero-shot-based, and watermark-based methods. These detectors are also classified as either white-box or black-box, depending on their access to LLM weights (Tang et al., 2023, Yang et al., 2023b). Watermarking-based methods typically fall under white-box detection, while proprietary models are restricted to black-box methods.

Featured-based Method. LLM-generated texts are typically less emotional and more objective than human-written texts (Guo et al., 2023). In comparison, human-authored texts are generally more coherent, while LLM-generated texts tend to repeat terms within a paragraph (Dugan et al., 2023). Similar to the problem of human authorship attribution, linguistic features such as phrasal verbs, coreference, part-of-speech (POS) tags, and named entity (NE) tags are also useful in distinguishing LLM-generated text (Fröhling and Zubiaga, 2021, Nguyen-Son et al., 2017, See et al., 2019). Feature-based methods are more explainable, but have drawbacks, such as poor generalizability of certain features across different domains and sampling methods.

Neural Network-Based Detectors. Neural network-based detectors, particularly those utilizing BERT, have proven effective in distinguishing between human-written texts and those generated by GPT-2 (Ippolito et al., 2020, Liu et al., 2019). Solaiman et al. (2019) fine-tuned the RoBERTa model with a dataset of GPT-2 outputs in open domain settings. Similarly, Guo et al. (2023) fine-tuned RoBERTa to detect ChatGPT-generated text. (Zhan et al., 2023) developed G3Detector by fine-tuning RoBERTa-large for the same purpose. Additionally, Chen et al. (2023a) introduced GPT-Sentinel, training both RoBERTa and T5 (Raffel et al., 2020) on their OpenGPTText dataset. In a different approach, Hu et al. (2023) created RADAR, which fine-tunes Vicuna 7B (Chiang et al., 2023) in a generative adversarial setting along with a paraphrase model. These efforts highlight the ongoing advancements in enhancing the detection of LLM-generated content using BERT-based models.

These detectors require retraining when encountering text from new LLMs to ensure reliable detection (Chakraborty et al., 2023b, Mitchell et al., 2023). Neural network-based detectors are vulnerable to adversarial and poisoning attacks (Goodfellow et al., 2015, Pu et al., 2023a, Wang et al., 2022). These detectors also face limitations such as overfitting to training data (Uchendu et al., 2020). The generalization ability of these detectors is critical, having been trained on various model families and tested on unseen models (Bhattacharjee et al., 2023, Pu et al., 2023b). Surrogate models, which are often small language models, are also applied to train classifiers (Mireshghallah et al., 2023, Verma et al., 2023).

Zero-Shot Detectors. Zero-shot detection methods are generally statistics-based, enabling the identification of LLM-generated text without additional training (Su et al., 2023b). Various statistical measures have been employed for detection, including entropy (Gehrmann et al., 2019, Lavergne et al., 2008), perplexity (Beresneva, 2016, Hans et al., 2024), average log-probability score (Solaiman et al., 2019), fluency (Holtzman et al., 2020), and Zipf's word frequency law (Piantadosi, 2014, Zipf, 2016)[2]. Additional methods

[2]The frequency of a word decreases as its rank in a frequency-ordered list increases.

leverage n-grams (Yang et al., 2023a), Uniform Information Density (UID) (Venkatraman et al., 2023), log rank information (Su et al., 2023b), and various linguistic features such as part-of-speech determiners, conjunctions, auxiliary relations, vocabulary, and emotional tone (Gehrmann et al., 2019, Joulin et al., 2017, Tang et al., 2023).

Various zero-shot detection methods provide distinct strategies to enhance both detection accuracy and efficiency. Gallé et al. (2021) introduced an unsupervised method that identifies over-appearance of repeated higher-order n-grams, distinguishing them from human-generated text. DetectGPT (Mitchell et al., 2023) relies on the observation that LLM-generated passages often fall into regions of negative curvature in log probability. (Bao et al., 2023) enhanced this approach and proposed Fast-DetectGPT, which increases efficiency by using conditional probability curvature. Additionally, some methods leverage LLMs themselves for text classification (Solaiman et al., 2019, Zellers et al., 2019). Different decoding methods are often applied to generate more diverse and less repetitive text, though these can also lead to hallucinations and less verifiable content (Guo et al., 2023, Shakeel and Jain, 2021). Fact-checking methods can mitigate these issues (Schuster et al., 2020, Zhong et al., 2020). Additionally, Krishna et al. (2024) developed a detector that uses information retrieval to store LLM outputs in a database and search for semantically similar content to identify LLM-generated text, though this method raises privacy concerns regarding the storage of user conversations.

Watermarking. Watermarking involves embedding specific patterns within text generated by LLMs, making them imperceptible to humans but detectable through specialized methods. (Kirchenbauer et al., 2023, Meral et al., 2009, Topkara et al., 2005, Zhao et al., 2023). By imprinting distinct patterns, watermarking enables the identification of LLM-generated text. Various methods include parsed syntactic tree structures (Atallah et al., 2001, Topkara et al., 2005), synonym tables (Jalil and Mirza, 2009), adversarial watermarking (Abdelnabi and Fritz, 2021), and context-aware lexical substitution (Yang et al., 2022). One notable approach is soft watermarking, proposed by Kirchenbauer et al. (2023), which partitions tokens into "green" and "red" lists to create patterns. A watermarked LLM samples tokens from the green list with high probability, determined by a pseudo-random generator seeded by its prefix token. The watermarking detector classifies passages with a high frequency of tokens from the green list as LLM-generated.

Further advancements have improved the robustness, efficiency, and stealthiness of watermarking methods (Hou et al., 2023, Wu et al., 2023a, Zhao et al., 2023). However, there is a trade-off between watermark effectiveness and text quality, as more reliable watermarks require more extensive text modifications (Sadasivan et al., 2023). Additionally, watermarking presents challenges for proprietary LLMs and third-party applications due to the necessity of accessing the language model's logits (Kirchenbauer et al., 2023). Watermark-based detection methodologies are also vulnerable to paraphrasing attacks (Krishna et al., 2024, Sadasivan et al., 2023).

3.3 Open Challenges

Detecting LLM-generated texts is challenging due to their high quality and ability to mimic human writing styles. LLMs can emulate human writing so closely that they pose significant challenges to traditional stylometric techniques. They can incorporate complex narrative structures and varied vocabularies, making it difficult to distinguish between human and LLM-generated texts. The rapid evolution of LLMs further complicates detection, as newer versions exhibit different stylistic traits, making detection models quickly obsolete (Chakraborty et al., 2023b). LLM-generated text detectors often struggle to generalize to unseen domains encountered during training (Li et al., 2023a, Pu et al., 2023b, Rodriguez et al., 2022) and tend to perform better on LLMs they were specifically trained on (Chakraborty et al., 2023b, Li et al., 2023a, Pu et al., 2023b).

Existing detectors also lack robustness to various factors, such as alternative decoding strategies (Ippolito et al., 2020), input sequence length (Solaiman et al., 2019), different prompts (Kumarage et al., 2023, Lu et al., 2023), repetition penalties (Fishchuk and Braun, 2023), and human edits (Zhang et al., 2024). Additionally, detectors are vulnerable to adversarial attacks, including homoglyph attacks (Gagiano et al., 2021, Macko et al., 2024), whitespace insertion (Cai and Cui, 2023), syntactic perturbations (Bhat and Parthasarathy, 2020), synonym replacement (Kulkarni et al., 2023), and paraphrasing (Becker et al., 2023, Krishna et al., 2024, Shi et al., 2024).

4 LLM-GENERATED TEXT ATTRIBUTION

Identifying whether a piece of text is generated by a specific LLM or a human is crucial. This distinction aids in tracing the origin of the text to ensure accountability, enhance transparency, and uphold ethical standards in information dissemination. If the content is harmful, misleading, or illegal, pinpointing the exact LLM responsible is essential for addressing ethical concerns and fulfilling legal obligations. LLM-generated text attribution builds upon techniques for LLM-generated text detection. Variations in model architecture (such as the number of layers and parameters), training methods (including pre-training and fine-tuning), and generation techniques (like sampling parameters) all subtly influence the characteristics of the generated texts (Munir et al., 2021).

4.1 Problem Definition

This attribution task extends beyond binary classification to handle multiple classes, increasing the complexity of LLM-generated text detection. The primary goal is to determine whether a given piece of text was created by a specific human or by one of several LLMs (Beigi et al., 2024, Chen et al., 2023b, He et al., 2023, Soto et al., 2024, Uchendu et al., 2020, Venkatraman et al., 2023). A sub-problem is to attribute the text solely to LLMs, also known as model sourcing (Yang et al., 2023a) or origin tracing (Li et al., 2023b).

4.2 Methodologies

Attributing texts to LLMs versus human writers involves recognizing inherent differences in their text generation capabilities. LLMs typically exhibit less diversity in word usage compared to humans (Dugan et al., 2023, Ippolito et al., 2020). LLMs can mimic a range of styles and tones, often masking their underlying characteristics. This ability to adapt makes attribution challenging, especially as LLMs rapidly evolve and their outputs change significantly over time (Guo et al., 2023). Additionally, LLMs may inadvertently reproduce snippets of their training data.

To simplify the classification process, it is common practice to group different human writers into a single category because humans exhibit a broader spectrum of writing styles and proficiency levels compared to machines (Uchendu et al., 2021). For example, classifications might include comparisons such as Human vs. ChatGPT, or Human vs. LLama (He et al., 2023, Uchendu et al., 2021). Some studies have formulated a 7-class classification that includes one human class and six LLM classes (He et al., 2023, Wang et al., 2024). Other approaches consider multiple human classes, albeit with a limited number. For instance, a 10-class classification might include seven human classes and three LLM classes (Tripto et al., 2023). This multiclass classification is often converted into a one-vs-rest classification for each label. Transformer-based models, such as BERT and RoBERTa, are fine-tuned on datasets containing both human-written and LLM-generated texts to conduct the attribution.

4.3 Open Challenges

Attributing texts generated by LLMs to specific humans or models presents a multi-class classification challenge. Variations in training data, model architecture, and fine-tuning processes contribute to the distinctive outputs of different LLMs, though these differences are often subtle and difficult to detect (Uchendu et al., 2021). Effective identification requires sophisticated methods to discern the unique signatures embedded in syntactic structures and lexical choices, which are influenced by specific training datasets. However, the proprietary nature of many LLMs restricts access to comparative data, posing significant hurdles. Additionally, the high degree of stylistic overlap among LLMs, especially those with similar architectures or trained on overlapping datasets, further complicates accurate classification. Continuous updates and fine-tuning of LLMs necessitate ongoing adjustments to attribution methodologies to account for evolving model characteristics (Wu et al., 2023b).

5 HUMAN-LLM CO-AUTHORED TEXT ATTRIBUTION

Besides creating text from scratch, LLMs are often used for extending sequences from human prompts. These perturbations, have diminished the effectiveness of existing text detection methods (Bhat and Parthasarathy, 2020). Identifying text that combines input from both human authors and LLMs presents unique challenges. Hybrid texts may originate as human-written content, with LLMs employed for conditional generation, making it difficult to clearly distinguish between the stylistic features of human and machine contributions. There are fewer studies on this task due to its difficulty, and existing research often makes simplifications.

5.1 Problem Definition

A human-LLM co-authored text, also known as mixed text (Zhang et al., 2024) or human-AI collaborative writing (Richburg et al., 2024), is a piece of writing that is partially created by a human and then revised or extended by LLMs, and vice versa. This task involves recognizing the nuances of multi-source authorship with fine-grained precision. Some studies categorize any text that is generated, modified, or extended by a machine as human-LLM co-authored text, treating it as LLM-generated. This simplifies the task to either LLM-generated Text Detection or LLM-generated Text Attribution (Crothers et al., 2023, Yang et al., 2023b). Other researchers handle LLM-revised human texts and human-revised LLM texts as a single category, alongside purely human-written and purely LLM-generated texts, approaching human-LLM co-authored text authorship attribution as a three-class classification problem (Richburg et al., 2024, Zhang et al., 2024). One variation of this task is to detect the boundary between human-written and LLM-generated text (Cutler et al., 2021, Wang et al., 2024).

5.2 Methodologies

Texts generated by LLMs are generally less emotional and more objective, often repeating terms within a paragraph. In contrast, human-authored texts tend to be more coherent and exhibit greater lexical diversity (Dugan et al., 2023, Guo et al., 2023, Zhang et al., 2024). Models like DNA-GPT (Yang et al., 2023a) and DetectGPT (Mitchell et al., 2023) utilized the T5 model (Raffel et al., 2020) to simulate scenarios where humans modify LLM-generated texts. MIXSET (Zhang et al., 2024) offers a more comprehensive dataset that includes text refined by LLMs through polishing, completion, and rewriting operations.

To effectively analyze and classify human-LLM co-authored texts versus those solely authored by humans or LLMs, feature-based methods from LLM-generated text detection, such as Log-likelihood (Solaiman et al., 2019), GLTR (Gehrmann et al., 2019), and log-rank (Mitchell et al., 2023), are adapted to this task. Additionally, neural network-based models like BERT (Ippolito et al., 2020), Radar (Hu et al., 2023), and GPT-sentinel (Chen et al., 2023a) can also be applied. The complexity of this task increases as users may employ multiple LLMs to compose different sections of an article, further blurring the lines between human and machine-generated content. Consequently, techniques used in earlier LLM detection need to evolve continuously. This ongoing evolution in detection strategies mirrors the increasing sophistication of LLM outputs and the collaborative nature of modern text creation.

5.3 Open Challenges

Authorship attribution involving human-written, LLM-generated, and human-LLM co-authored pieces presents varying degrees of complexity, requiring distinct analytical approaches to accurately identify and differentiate the contributions of each author. For texts authored entirely by humans or LLMs, stylometric techniques can be effectively utilized. Human-authored texts often feature unique stylistic nuances, such as variable sentence structures and emotive language (Zhang et al., 2024). In contrast, LLM-generated texts typically exhibit consistent syntax and a broader vocabulary (Guo et al., 2023). Feature-based methods used in LLM-generated text detection can be adapted to classify texts by identifying these distinct patterns, thereby attributing texts to their correct source.

Analyzing and classifying texts co-authored by humans and LLMs presents a significant challenge due to the blending of human and machine stylistic features. These texts often start as human drafts and are later extended or revised by LLMs, or the process might occur in reverse. This integration of styles creates a hybrid form that makes it difficult to distinguish distinct authorial markers, thereby complicating the attribution process.

Human-LLM co-authored texts pose a more intricate challenge due to the blending of stylistic and linguistic elements from both human authors and LLMs. These texts may begin as human drafts later extended or revised by LLMs, or vice versa, resulting in an integration of styles that obscures distinct authorial markers (Liu et al., 2023a). Advanced techniques are required to dissect these integrations, identifying where and how LLM contributions intersect with human input (Wang et al., 2024). This involves detecting subtle shifts in style and contextual cues that indicate the extent of LLM involvement, allowing for accurate segmentation and attribution of authorship within hybrid documents.

6 RESOURCES AND EVALUATION METRICS

This section provides an in-depth examination of widely used benchmarks and datasets in authorship attribution research, along with guidelines for selecting appropriate ones. These resources range from purely human-written texts to those generated by LLMs and human-LLM co-authored texts. This diversity is crucial for training and evaluating models across various tasks.

Traditional datasets focus exclusively on texts written by humans, while modern datasets include LLM-generated text, addressing the need to detect and attribute text produced by LLMs. Additionally, this section covers commercial and open-source detectors commonly used to identify machine-generated text. Lastly, we summarize the common evaluation metrics employed in this field.

6.1 Benchmarks and Datasets

Authorship datasets encompass a wide range of sources, from formal literature to informal online communications, highlighting the increasing significance of user-generated content on social media. Human authorship datasets typically include author identifiers and ideally contain multiple texts for each author. Manually collecting data for large datasets is time-consuming and costly, motivating researchers to utilize web data sources such as Wikipedia and Reddit. In contrast, custom datasets for LLM-generated text are easier and more affordable to create, and they are often built alongside human-written text to maintain a similar domain and format.

A general guideline for selecting and constructing datasets involves incorporating variations in domain, model architecture, and decoding strategies. Addressing class imbalance is crucial, as LLM-generated and human-written texts are often disproportionate. For human authorship data, it is recommended to choose content created before the widespread use of LLMs (GPT-3 (Brown et al., 2020) was released in June 2020, and ChatGPT followed in November 2022) to ensure that the texts were predominantly human-written.

Factors influencing the performance of existing authorship attribution models include the size of the training text Factors influencing the performance of existing authorship attribution models include the size of the training text (Hirst and Feiguina, 2007, Marton et al., 2005), the number of candidate authors (Koppel et al., 2006), and the imbalanced distribution of training texts among the candidate authors (Stamatatos, 2008). The availability of digital text in formats such as tweets, blogs, and articles has exponentially increased, providing more training data to accelerate the development of authorship attribution. However, the rapid growth of online communication has also led to shifts in writing behavior, resulting in shorter, fragmented, and less coherent social media tweets and text messages. For example, tweets are limited to 280 characters, whereas legal judgment documents contain thousands of words (Seroussi et al., 2011). The challenge in social media stems from the brief nature of posts and a large pool of potential authors, making the attribution of short documents particularly difficult (Aborisade and Anwar, 2018, Seroussi et al., 2014, Theophilo et al., 2021).

Table 1 provides a comprehensive overview of widely used benchmarks and datasets. These datasets are characterized by statistics including domain, size, word length, language, and the LLMs used to generate texts. All listed datasets support LLM-generated text detection (Problem 2). However, fewer support LLM-generated text attribution (Problem 3) and Human-LLM Co-authored Text Attribution (Problem 4). These benchmarks often originate from human-written datasets like XSum (Narayan et al., 2018), OpenWebText (Gokaslan and Cohen, 2019), and Wikipedia. Since Problems 2, 3, and 4 frequently treat human-written text as a single category for simplicity, rather than identifying individual authors as in Problem 1, many of these datasets are unsuitable for Problem 1, which requires unique author identification. Therefore, the representative datasets for human authorship are summarized as follows:

- Amazon Review (Ni et al., 2019): Featuring reviews with ratings, text, votes, product metadata, and links, this dataset provides a comprehensive view of consumer opinions, ideal for commercial authorship attribution studies.
- Aston 100 Idiolects Corpus (Heini and Kredens, 2021): Comprising emails, essays, text messages, and business memos from 100 individuals (ages 18–22, native English speakers), this corpus provides a broad spectrum of text types for analyzing both content and stylistic features.
- Blog Authorship Corpus (Schler et al., 2006): Contains over 680,000 posts from more than 19,000 authors, with an average of 35 posts per author. The texts, averaging 79 tokens, are informal and conversational, typical of blog posts.
- Deceptive Opinion Spam (Ott et al., 2011): Includes 400 genuine and 400 deceptive hotel reviews, with deceptive reviews generated using Amazon Mechanical Turk, useful for studying the nuances of fake versus real reviews.
- Enron Email (Klimt and Yang, 2004): Includes around 500,000 messages from 160 employees, offering long texts and high text-per-author variance, making it ideal for studying corporate communication styles.
- Fanfiction: Collected from fanfiction.net, this dataset includes fan-written fiction (Bischoff et al., 2020, Kestemont et al., 2021), providing insights into creative writing and authorship attribution in fictional narratives.
- IMDb1M (Seroussi et al., 2014): Features over 270,000 movie reviews by 22,000 authors, with an average of 12.3 texts per author and an average text length of 121 tokens, suitable for analyzing shorter, user-generated content.
- Pushshift Reddit (Baumgartner et al., 2020): This dataset comprises posts and comments from various subreddits, covering diverse topics and writing styles, making it suitable for analyzing informal online discourse.
- PAN (Bevendorff et al., 2022, Kestemont et al., 2021): Offered by PAN workshops for benchmarking authorship

Name	Domain	Size	Length	Language	Model	P2	P3	P4
TuringBench (Uchendu et al., 2021)	News	168,612 (5.2%)	100 to 400 words	en	GPT-1,2,3, GROVER, CTRL, XLM, XLNET, FAIR, TRANSFORMER-XL, PPLM	✓	✓	
TweepFake (Fagni et al., 2021)	Social media	25,572 (50.0%)	less than 280 characters	en	GPT-2, RNN, Markov, LSTM, CharRNN	✓		
ArguGPT (Liu et al., 2023b)	Academic essays	8,153 (49.5%)	300 words on average	en	GPT2-Xl, text-babbage-001, text-curie-001, davinci-001,002,003, GPT-3.5-Turbo	✓		
AuTexTification (Sarvazyan et al., 2023)	Tweets, reviews, news, legal, and how-to articles	163,306 (42.5%)	20 to 100 tokens	en, es	BLOOM, GPT-3	✓	✓	
CHEAT (Yu et al., 2023)	Academic paper abstracts	50,699 (30.4%)	163.9 words on average	en	ChatGPT	✓		
GPABench2 (Liu et al., 2023a)	Academic paper abstracts	2.385M (6.3%)	70 to 350 words	en	ChatGPT	✓		✓
Ghostbuster (Verma et al., 2023)	News, student essays, creative writing	23,091 (87.0%)	77 to 559 (median words per document)	en	ChatGPT, Claude	✓		
HC3 (Guo et al., 2023)	Reddit, Wikipedia, medicine, finance	125,230 (64.5%)	25 to 254 words	en, zh	ChatGPT	✓		
HC3 Plus (Su et al., 2023a)	News, social media	214,498	N/A	en, zh	ChatGPT	✓		
HC-Var (Xu et al., 2023)	News, reviews, essays, QA	144k (68.8%)	50 to 200 words	en	ChatGPT	✓		
HANSEN (Tripto et al., 2023)	Transcripts of speech (spoken text), statements (written text)	535k (96.1%)	less than 1k tokens	en	ChatGPT, PaLM2, Vicuna-13B	✓	✓	
M4 (Wang et al., 2023)	Wikipedia, WikiHow, Reddit, QA, news, paper abstracts, peer reviews	147,895 (24.2%)	more than 1k characters	ar, bg, en, id, ru, ur, zh	davinci-003, ChatGPT, GPT-4, Cohere, Dolly2, BLOOMz	✓		
MGTBench (He et al., 2023)	News, student essays, creative writing	21k (14.3%)	1 to 500 words	en	ChatGPT, ChatGLM, Dolly, GPT4All, StableLM, Claude	✓	✓	
MULTITuDE (Macko et al., 2023)	News	74,081 (10.8%)	200 to 512 tokens	ar, ca, cs, de, en, es, nl, pt, ru, uk, zh	GPT-3,4, ChatGPT, Llama-65B, Alpaca-LoRa-30B, Vicuna-13B, OPT-66B, OPT-IML-Max-1.3B	✓		
OpenGPTText (Chen et al., 2023a)	OpenWebText	58,790 (50.0%)	less than 2k words	en	ChatGPT	✓		
OpenLLMText (Chen et al., 2023b)	OpenWebText	344,530 (20%)	512 tokens	en	ChatGPT, PaLM, Llama, GPT2-XL	✓	✓	
Scientic Paper (Mosca et al., 2023)	Scientific papers	29k (55.2%)	900 tokens on average	en	SCIgen, GPT-2,3, ChatGPT, Galactica	✓		
RAID (Dugan et al., 2024)	News, Wikipedia, paper abstracts, recipes, Reddit, poems, book summaries, movie reviews	523,985 (2.9%)	323 tokens on average	cs, de, en	GPT-2,3,4, ChatGPT, Mistral-7B, MPT-30B, Llama2-70B, Cohere command and chat	✓		
M4GT-Bench (Wang et al., 2024)	Wikipedia, Wikihow, Reddit, arXiv abstracts, academic paper reviews, student essays	5,368,998 (96.6%)	more than 50 characters	ar, bg, de, en, id, it, ru, ur, zh	ChatGPT, davinci-003, GPT-4, Cohere, Dolly-v2, BLOOMz	✓	✓	✓
MAGE (Li et al., 2023a)	Reddit, reviews, news, QA, story writing, Wikipedia, academic paper abstracts	448,459 (34.4%)	263 words on average	en	GPT, Llama, GLM-130B, FLAN-T5 OPT, T0, BLOOM-7B1, GPT-J-6B, GPT-NeoX-2	✓		
MIXSET (Zhang et al., 2024)	Email, news, game reviews, academic paper abstracts, speeches, blogs	3.6k (16.7%)	50 to 250 words	en	GPT-4, Llama2	✓		✓

Table 1: Summary of Authorship Attribution Datasets and Benchmarks with LLM-Generated Text. Size is shown as the sum of LLM-generated and human-written texts (with the percentage of human-written texts in parentheses). Language is displayed using the two-letter ISO 639 abbreviation. Columns P2, P3, and P4 indicate whether the dataset supports problems described in Problem 2, 3, and 4, respectively.

Detector	Pricing Plan	API	Website
GPTZero	150k words at $10/month, 10k words for free per month	Yes	https://gptzero.me/
ZeroGPT	100k characters for $9.99, 15k characters for free	Yes	https://www.zerogpt.com/
Sapling	50k characters for $25, 2k characters for free	Yes	https://sapling.ai/ai-content-detector
Originality.AI	200k words at $14.95/month	Yes	https://originality.ai/
CopyLeaks	300k words at $7.99/month	Yes	https://copyleaks.com/ai-content-detector
Winston	80k words at $12/month	Yes	https://gowinston.ai/
GPT Radar	$0.02/100 tokens	N/A	https://gptradar.com/
Turnitin's AI detector	License required	N/A	https://www.turnitin.com/solutions/topics/ai-writing/ai-detector/
GPT-2 Output Detector	Free	N/A	https://github.com/openai/gpt-2-output-dataset/tree/master/detector
Crossplag	Free	N/A	https://crossplag.com/ai-content-detector/
CatchGPT	Free	N/A	https://www.catchgpt.ai/
Quil.org	Free	N/A	https://aiwritingcheck.org/
Scribbr	Free	N/A	https://www.scribbr.com/ai-detector/
Draft Goal	Free	N/A	https://detector.dng.ai/
Writefull	Free	Yes	https://x.writefull.com/gpt-detector
Phrasly	Free	Yes	https://phrasly.ai/ai-detector
Writer	Free	Yes	https://writer.com/ai-content-detector/

Table 2: Overview of LLM-Generated Text Detectors.

attribution and verification models and are used in various authorship attribution competitions.

- VALLA (Tyo et al., 2022): Designed for benchmarking authorship attribution models, VALLA standardizes a range of texts across various genres and writing styles.

Table 2 summarizes popular commercial LLM-generated text detectors. These detectors are primarily designed for LLM-generated text detection (Problem 2). Some detectors like GPTZero (Tian and Cui, 2023) can also detect human-LLM co-authored text and identify portions likely to be LLM-generated, addressing the boundary detection sub-problem of Problem 4 (Cutler et al., 2021). Although many of these detectors claim over 99% accuracy in detecting LLM-generated text, few are tested on shared benchmark datasets. Despite their high accuracy claims, many detectors suffer from high false positive rates, which could falsely accusing individuals of plagiarism and undermining credibility and trust in genuine authors. Additionally, these detectors often lack robustness against variations in sampling strategies, adversarial attacks, and unseen domains and language models (Dugan et al., 2024).

6.2 Evaluation Metrics

Evaluation metrics such as F1 score and AUCROC are essential for quantifying the performance of authorship models, providing a standardized means to assess and compare the effectiveness of different authorship attribution approaches. As in other classification tasks, existing studies predominantly use the Area Under the Receiver Operating Characteristic (AUCROC) and F1 score to evaluate attribution algorithms. In human authorship attribution, where there are a large number of candidate authors, retrieval metrics like Mean Reciprocal Rank (MRR) and recall-at-k are used (Rivera-Soto et al., 2021). Additionally, Self-BLEU and perplexity are useful metrics, with a lower Self-BLEU score indicating higher textual diversity (Zhang et al., 2024). Common evaluation metrics include:

- Accuracy: Measures the proportion of correctly identified authors but can be misleading in imbalanced datasets.
- Precision, Recall, and F1-Score: Crucial in imbalanced datasets. Precision indicates the relevance of identified instances, recall measures the ability to identify all relevant instances, and the F1-Score balances both.
- Area Under the Receiver Operating Characteristic Curve (AUCROC): Represents the trade-off between true positive rates and false positive rates, where higher values indicate better performance.
- False Positive Rate (FPR) and False Negative Rate (FNR): Critical for minimizing misclassification, with FPR measuring incorrect classification of human texts as LLM-generated and FNR the reverse.
- Recall-at-k: measure the probability that the correct author appears among the top k results when ranking targets by cosine similarity to a query text.
- Mean Absolute Error (MAE): Used to evaluate the performance of human-machine text boundary detection. It measures the average absolute difference between the predicted position index and the actual change point.

7 OPPORTUNITIES AND FUTURE DIRECTIONS

This section explores future directions in the field of authorship attribution, focusing on leveraging the potential of LLMs while addressing associated challenges. Future efforts should aim for finer granularity in authorship attribution, leveraging LLM capabilities, improving generalization, enhancing explainability, preventing misuse, developing standardized benchmarks, and integrating interdisciplinary perspectives to enrich the field.

7.1 Finer Granularity

Current authorship attribution methods often face limitations when handling a more extensive range of candidate human authors or LLMs, presenting opportunities for future research. For instance, existing approaches for LLM-generated Text Attribution typically manage only a limited number of authors or models, which restricts their applicability in real-world scenarios where the pool of potential authors or models can be vast. Previous studies have often oversimplified the problem by categorizing all human-written text as a single category (He et al., 2023, Uchendu et al., 2021). This approach ignores the diversity among human authors and fails to leverage the rich set of characteristics that distinguish individual writing styles. Future work can build upon traditional research on human authorship to develop methods capable of attributing human-written text to individual authors even within the context of LLM-generated content. This refinement will improve the accuracy and utility of authorship attribution models, especially in mixed datasets containing both human-written and LLM-generated texts.

Similarly, for Human-LLM Co-authored Text Attribution, there is a need to attribute text more precisely to individual human authors or specific LLMs. Current work simplifies human-written text, LLM-generated content, and text co-authored by humans and LLMs into three broad categories, without differentiating between individual human authors and specific LLMs (Richburg et al., 2024, Zhang et al., 2024). This approach overlooks the nuanced contributions of each author or model. By improving the granularity of attribution, future models can better distinguish between various human authors and LLMs, thus increasing the practicality and reliability of authorship attribution tools. Such advancements would be particularly valuable in collaborative environments where multiple human authors and LLMs contribute to a single body of work, enabling clearer recognition of each contributor's role.

7.2 Generalization

This subsection examines the applicability of current methodologies across varying LLMs, domains, genres, and languages. Domains refer to broad areas of knowledge or topics, while genres refer to specific styles or forms of writing within any domain. Domain generalization poses significant challenges due to variability in vocabularies, syntax, and styles across different subjects, complicating accurate authorship attribution. Attribution performance tends to drop when known and query texts differ in topic or genre (Altakrori et al., 2021). Models like BERT and RoBERTa have shown limitations in cross-domain tasks (Barlas and Stamatatos, 2020, Huertas-Tato et al., 2022), and adapting models to new domains remains difficult due to factors like dataset size variability and writing styles. Traditional methods focused on identifying less topic-dependent features such as function words and part-of-speech n-grams (Madigan et al., 2005b, Menon and Choi, 2011), while recent approaches highlight the importance of training more powerful transformer-based models (Rivera-Soto et al., 2021) and techniques such as adversarial training (Ben-David et al., 2010, Ganin et al., 2016, Li et al., 2017).

Genre generalization involves adapting to different writing styles, such as fiction, nonfiction, and poetry, each with unique features. Authors' adaptation to various genres dilutes their identifiable stylistic traits, complicating attribution. Similarly, distinguishing between human-written and LLM-generated text in different genres requires identifying genre-specific inconsistencies. The diversity of genres demands flexible models capable of understanding various narrative structures, tones, and stylistic elements. Adapting models to handle genre variations necessitates more advanced and flexible approaches for effective generalization. Current authorship attribution models also struggle with out-of-distribution issues when faced with languages and LLMs not encountered during training, leading to decreased accuracy and reliability (Koppel et al., 2005, Wu et al., 2023b). Addressing this generalization problem is crucial for developing robust models that can handle diverse and evolving linguistic and model landscapes.

Improving generalization can potentially be achieved through several strategies. First, leveraging transfer learning by pre-training on large, diverse datasets and fine-tuning on specific domains enhances adaptability and performance (Barlas and Stamatatos, 2020, Rodriguez et al., 2022). Second, developing domain and genre-invariant features would improve robustness by focusing on core stylistic elements (Argamon et al., 2003). Third, employing hybrid models and ensemble methods that integrate domain-specific knowledge can optimize prediction accuracy by drawing on the strengths of individual models (Bacciu et al., 2019). Additionally, incorporating contextual factors like the writing environment or intended audience, alongside data augmentation techniques, can bolster generalization. Finally, improving attribution in multilingual contexts enables models to operate effectively across various languages (Chen et al., 2022, Shamardina et al., 2022). As detectors could be biased against non-native English writers (Liang et al., 2023), enhancing multilingual generalization is crucial for fairness. Collectively, these approaches foster robust and adaptable models equipped to handle diverse styles and contexts.

7.3 Explainability

Improving explainability is crucial for ensuring transparency and trust in authorship attribution models, especially as they become increasingly integrated into fields such as forensic investigation, academia, and journalism. Developing explainable AI techniques specifically for authorship attribution can lead to methodologies where the reasoning behind attributions is clear and understandable. This transparency is particularly important when authorship attribution is used as evidence in legal proceedings (Chaski, 2005, Rocha et al., 2016). Traditional attribution methods, which rely on stylistic and linguistic features to identify an author, often struggle to distinguish between human-authored texts and those generated by LLMs, which can adeptly replicate these features. This challenge underscores the need for enhanced methodologies that not only differentiate between human and AI-generated content but also explore how LLMs emulate specific authorial styles (Boenninghoff et al., 2019a, Danilevsky et al., 2020).

Despite attempts such as analyzing internal attention weights or employing interpretation tools and visualization techniques (Wallace et al., 2019), word-level explanations are insufficient. The challenge remains to provide higher-level explainability that aligns with human cognitive processes (Rudin, 2019). Advances in this area may include leveraging discourse-level relations and training models with human explanations for common sense reasoning to

improve the explanatory depth of model-generated attributions (Rajani et al., 2019). For instance, Kowalczyk et al. (2022) detected GPT-2-generated fake reviews using Shapley Additive Explanations (SHAP) (Lundberg and Lee, 2017).

7.4 Misuse Prevention

Future research should focus on refining existing authorship attribution methods to detect and prevent malicious activities such as generating misinformation, plagiarism, and propaganda (Goldstein et al., 2023, Hazell, 2023, Lund et al., 2023, Spitale et al., 2023). These methods analyze stylistic features to detect discrepancies in claimed authorship and trace the origins of content, thereby identifying suspicious texts. For plagiarism, attribution models can compare writing styles to a database of known authors. In combating misinformation and propaganda, these models could identify and flag content patterns typical of known propagandists.

To ensure effectiveness in real-world tasks, authorship attribution models should be robust against out-of-domain data and adversarial attacks. Adversarial attacks, such as alternative spellings, article deletion, paragraph additions, case changes, zero-width spaces, whitespace manipulation, homoglyphs, number swaps, misspellings, paraphrasing, and synonym substitution, have been shown to effectively degrade detector performance (Dugan et al., 2024). Diversifying training data with various writing styles and topics could improve out-of-domain robustness. Adversarial robustness could be achieved through adversarial training and employing ensemble methods to build resilience against intentional manipulations.

7.5 Leveraging LLM Capabilities

Leveraging LLMs can enhance both traditional feature-based stylometry methods and LLM-based approaches. By integrating LLMs with existing methods, researchers can gain deeper insights into stylistic nuances, improving the robustness of authorship detection across various textual genres and lengths. The increasingly large context length of LLMs enables in-context learning (ICL) (Brown et al., 2020) by incorporating more documents, enhancing the model's ability to capture intricate writing patterns.

Another promising approach is Retrieval-Augmented Generation (RAG) (Lewis et al., 2020). RAG can enhance authorship attribution by retrieving additional documents for each author, thereby assisting in generating more contextually accurate results. Moreover, leveraging LLMs for data augmentation and synthetic data generation can create diverse training datasets, which in turn improves the generalization of attribution models (Albalak et al., 2024).

Combining text detection with other modalities, such as images and videos, can potentially enhance authorship attribution. Multi-modal analysis offers a more comprehensive understanding of the content and its origins by integrating diverse data types. This holistic approach not only improves the attribution process but also fosters the development of innovative methodologies that are better adapted to the evolving nature of digital content.

7.6 Developing Standardized Benchmarks

The diversity in datasets and evaluation metrics currently poses a challenge to the consistent and fair evaluation of different authorship attribution methods. To advance the field, it is crucial to establish benchmarks that encompass a broad spectrum of text types and sources, including those that are human-authored, LLM-generated, and human-LLM co-authored. These benchmarks should integrate a diverse range of text corpora, representing various sources, domains, text lengths, and languages, to accurately reflect real-world applications. Moreover, unified and comprehensive evaluation metrics are essential for ensuring consistent and transparent measures of attribution accuracy, robustness, explainability, and computational efficiency, thereby enabling fair comparisons across different models. Additionally, benchmarks should include datasets that combine human-written and machine-generated content from various authors and LLMs, simulating realistic tasks and testing the robustness of attribution models.

To ensure ongoing relevance and challenge for attribution methods, benchmarks must be regularly updated to include new types of LLMs and detectors. By developing and adopting standardized benchmarks, the research community can foster more rigorous, reproducible, and comparable studies. This will ultimately drive advancements in authorship attribution methodologies and applications. These standardized benchmarks would serve as a foundation for the systematic evaluation of attribution techniques, promoting innovation and progress in addressing the complexities of authorship attribution in a rapidly evolving digital landscape.

7.7 Integrating Interdisciplinary Perspectives

Authorship attribution is inherently multidisciplinary, encompassing elements of linguistics, computer science, forensic science, and psychology (Stamatatos, 2009). Future research should continue fostering collaboration across these fields to integrate diverse perspectives and methodologies. This integrative approach can lead to innovative solutions and a deeper understanding of the challenges and potential of authorship attribution. Combining insights from various disciplines can foster the creation of holistic attribution models that account for both the intricacies of human language and the complexities of LLM-generated texts. Such collaboration could also spearhead initiatives to standardize evaluation metrics for authorship attribution tools, ensuring their effectiveness across diverse contexts and compliance with ethical standards.

Linguistics can dissect textual structures and stylistic nuances, identifying unique linguistic fingerprints of authors. It explores novel features that enhance robustness across domains and improve explainability. Forensic science contributes through technological tools and methodologies, enabling precise examination of physical and digital texts. Psychology, particularly psycholinguistics, provides insights into how the brain processes function words and grammatical markers distinctively from lexical content words, revealing correlations with socio-cultural categories such as gender, age, and native language (Chambers et al., 2013, Nerbonne, 2014, Seals and Shalin, 2023), which are pivotal in understanding identity and social affiliations (Argamon et al., 2009). The Linguistic Inquiry and Word Count (LIWC) tool exemplifies how automated text analysis can use more than 100 psychological dimensions to analyze word use, reflecting distinct language variations among different groups in specific genres and languages (Dudău and Sava, 2021, Goldstein-Stewart et al., 2009, Pennebaker et al., 2015).

Combining these interdisciplinary perspectives enhances our ability to distinguish between human and machine-generated texts,

addressing the emerging challenges posed by sophisticated language models. By integrating linguistic theory with advanced computational techniques, forensic methodologies, and psychological insights, researchers can develop more comprehensive and nuanced authorship attribution frameworks. These frameworks will be better equipped to handle the diverse range of writing styles and contexts, ultimately leading to more accurate and reliable attribution outcomes. Furthermore, interdisciplinary collaboration can drive the development of ethical guidelines and best practices, ensuring that authorship attribution is conducted responsibly and with respect for individuals' privacy and rights.

8 ETHICAL AND PRIVACY CONCERNS

In the evolving landscape of authorship attribution, it is crucial to prioritize ethical considerations to safeguard privacy, integrity, and the rightful ownership of content. The attribution of text to specific authors or models raises significant ethical and privacy issues. Misattribution can lead to wrongful accusations or misinterpretation of an author's intent (Lund et al., 2023). Additionally, the use of attribution technologies must balance the need for accountability with respect for individuals' privacy and the potential for misuse in surveilling or censoring content.

Authorship attribution techniques are essential in digital forensics, cybersecurity, and plagiarism detection. However, the potential to reveal the identities of anonymous authors presents significant ethical challenges. Applications such as linking user accounts across platforms and identifying compromised accounts raise privacy concerns and ethical questions about surveillance and profiling individuals based on their writing style.

The use of authorship attribution methods must be carefully managed to protect individual privacy and adhere to ethical standards, particularly in sensitive areas like journalism, political dissent, and corporate whistle-blowing (Sison et al., 2023). Ensuring that these methods are not used to undermine privacy rights or expose individuals to risks without their consent is essential. Despite existing measures to prohibit the unethical use of LLMs, these restrictions could be evaded through prompt engineering and jail-breaking, posing risks of phishing and fraud scams.

Furthermore, the increasing difficulty in distinguishing between human and LLM-generated content raises concerns about intellectual property, plagiarism, and accountability. Accurate attribution is crucial for maintaining academic and creative integrity, yet tools and methods for achieving this must evolve rapidly to keep pace with technological advancements. The deployment of LLMs in generating content across various domains—from journalism to literature—necessitates a rethinking of authorship norms and the legal frameworks governing creative works.

9 CONCLUSION

The field of authorship attribution is experiencing both unprecedented challenges and remarkable opportunities with the advent of LLMs. Whether the objective is to identify human authors, differentiate between human and machine-generated texts, attribute texts to specific LLMs, or manage the complexities of human-LLM co-authored texts, ongoing innovation is imperative. Effectively addressing these multifaceted issues requires interdisciplinary approaches and collaborative efforts among researchers. This review explores various problems within authorship attribution, offering a comprehensive comparison of methodologies and datasets. By integrating robustness, explainability, and interdisciplinary perspectives, the insights gained are not only accurate but also socially relevant and trustworthy. The review highlights the strengths and limitations of current approaches, identifies key open problems, and outlines future research directions. This holistic analysis equips researchers and practitioners with the knowledge necessary to navigate the evolving landscape of authorship attribution, emphasizing critical areas for future research and development.

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